GGG-H-142G
August 26, 1969
SUPERSEDING
Int. Fed. Spec. GGG-H-00142f (NAVY-Ships)
June 10, 1968 and
Fed. Spec. GGG-H-142e
June 6, 1967

FEDERAL SPECIFICATION

HELMET, CONSTRUCTION WORKER'S

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for two types of safety helmets for protection of the wearer's head against impact, penetration, and electrical shock.

1.2 Classification.

1.2.1 Types. The helmets shall be furnished in the following types, as specified (see 6.2).

Type I Hat (full brim).
Type II Cap (brimless with peak).

1.2.2 Colors. Helmets shall be furnished in the colors listed in 3.5, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 406 - Plastics: Methods of Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

FSC 8415

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Military Specification:

MIL-L-10547 - Liners, Case, and Sheet Overwrap; Water-Vaporproof or Water-proof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available from reference libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The helmets furnished under this specification shall be products which are qualified for listing on the applicable products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Components. Each helmet shall consist of a shell, suspension (headband and crown straps forming an internal cradle for supporting the helmet on the wearer's head), a sweatband, and when specified, shall include an adjustable chinstrap, nape strap or winter liner (see 6.2).

3.3 Materials. Materials used in the fabrication of the helmet shells furnished under this specification shall be at the option of the manufacturer. All materials coming in contact with the wearer's head shall be non-irritating to normal skin.

3.4 Construction.

3.4.1 Shell. The shell shall be of one-piece seamless construction, dome shaped to provided the inside clearances specified herein.

3.4.1.1 Metal parts. Metal parts shall not impair the electrical protection which the shell provides for the wearer. They shall be so placed that no part of the worker's head is likely to be scratched by a blow on the brim of the helmet which would cause the helmet to slide down on the ears or forehead. Metal parts shall be rust resistant.

3.4.1.2 Brim.

3.4.1.2.1 Type I, full brim. The brim for type I helmets shall be an integral extension of the shell dome, and shall be continuous around the helmet. The width of the brim shall be not less than 1-1/2 inches and not more than 3 inches measured from the inside edge of the helmet except that the width at the front may taper to not less than 1-1/4 inches. With the helmet held in a horizontal position, the brim shall slope downward in a straight line at an angle of not less than 15 degrees nor more than 37 degrees from the horizontal.

3.4.1.2.2 Type II, brimless with peak. The peak for type II helmets shall extend forward from the crown not less than 1-1/2 inches, and shall be not less than 5-1/2 inches wide at the narrowest width. Type II helmets shall not be made by cutting the brim from type I helmets.

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3.4.2 Headbands. Headbands shall be adjustable and shall permit adjustment to all sizes within the range marked on helmet or headband. The headband (which may be a replaceable component) shall be securely attached either to the helmet shell or to the crown straps for assuring its proper position in the helmet. The manner of the attachment shall not interfere with the adjustments of the crown straps.

3.4.2.1 Sweatband. Sweatbands of the removable-replaceable type shall be provided or if sweatband is integral with headband, the headband shall be a replaceable component of the suspension assembly. The sweatband shall cover not less than the forehead portion of the headband and shall be at least as wide as the headband.

3.4.2.2 Adjustment range. Headbands shall be adjustable in 1/8 size increments from at least size 6-3/4 to 7-1/2, inclusive. The sizes shall be such that the circumference of the inside of the headband for size 7 is 22 inches, with a 3/8 inch increment for each 1/8 size above and below size 7. Tolerances of circumferential measurements shall be ± 1/8 inch.

3.4.2.3 Headband clearance. As determined by the method described in 4.5.7 the clearance between the inside of the shell and the outside of the headband shall be not less than 1/4 inch nor more than 3/4 inch.

3.4.3 Crown straps. Crown straps shall be not less than 1/2 inch in width. These straps, when adjusted and properly interlaced, shall form a cradle or hammock for supporting the helmet on the wearer's head and holding the helmet away from the head, thus acting as a shock absorber when the helmet is struck. The number of straps and their location in the helmet shall be suitable for the above purpose. Crown straps shall be so designed that the clearance between the standard head form and the shell of the hat or cap cannot be adjusted to less than 1-1/4 inches minimum crown clearance requirement permitted in 4.5.3.4(b).

3.4.4 Accessories. If chinstraps, nape straps or winter liners are specified in the contract or order (see 6.2), they shall conform to the following requirements:

3.4.4.1 Chinstrap. The chinstrap shall be adjustable and made of closely woven webbing, leather, elastic webbing combination, or plastic, of not less than 1/2 inch in width.

3.4.4.2 Winter liner. The winter liner shall be designed to cover the skull, neck and ears. The skull cap, part of the winter liner, shall be attached to the helmet cradle or to the helmet in such a manner that it can be readily removed and reattached. The earlug and neck part may be made either in one piece with the skull cap or attachable to it by tabs or a zipper, as specified. The neck and earlug shall be drawn close to the face by a lace or a chinstrap. The winter liner shall consist of fabric, plastic or any suitable material; colored material shall be colorfast. The outer surface may be water repellant.

3.4.4.3 Nape strap. The nape strap shall be adjustable and made of closely woven webbing, leather, elastic cotton webbing combination or plastic, of not less than 1/2 inch in width. (NOTE: A separate adjustable nape strap shall not be furnished when headband is equipped with integral contoured nape strap).

3.5 Colors. The color of the helmets shall be as specified and shall conform to Fed. Std. No. 595 (see 6.2) Department of Defense purchases shall be restricted to the following standard colors:

Color	Color Number (Fed. Std. No. 595)*
Maroon	10075
Terra Cotta	10233
Red	11105
Orange	12197
Yellow	13538
Green	14260
Blue	15177
Gray	16492
Black	17038
White	17875

* Individual (3 by 5 inch) color chips are available from General Services Administration, Business Service Center, Washington, D.C. 20407.

3.5.1 Color application. The color shall be provided by the use of high-quality pigment, pigmented lacquers, alkyd enamel or polyester finish. Injection molded shells shall be pigmented with applicable color during molding.

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3.6 Physical requirements.

3.6.1 Insulation resistance. The helmet shall not show a leakage current in excess of 3 milliamperes when tested as specified in 4.5.2.

3.6.2 Impact resistance. When tested in accordance with 4.5.3, helmets shall not transmit an average force of more than 850 pounds. Individual specimens shall not transmit a force greater than 1,000 pounds. Disengagement of, deformation of, or damage to the shell or component parts shall be allowed. The shell shall be permitted to touch the standard head form provided the transmitted force requirement is complied with.

3.6.3 Penetration strength. The helmet shall be neither dented nor pierced for more than 3/8 inch, nor shall the helmet be forced down over the standard head form, nor shall the straps pull out or break when tested as specified in 4.5.4.

3.6.4 Flammability. The thinnest section of the shell shall not burn at a rate greater than 3 inches per minute when tested as specified in 4.5.5.

3.6.5 Water absorption. The shell material of the helmet shall absorb not more than 5 percent of water when tested as specified in 4.5.6.

3.6.6 Weight. The weight of each complete helmet without winter liner and chinstrap or other accessories shall not exceed 15 ounces (see 4.5.8).

3.7 Interchangeability. The suspension shall be easily removable and replaceable and shall be interchangeable among shells of the same size and model furnished by the same manufacture (see 6.2).

3.8 Instructions. Instructions which explain the propoer method of wearing and adjusting the helmet shall be furnished with each helmet. They shall be stowed in the helmet so as to remain where placed until intentionally removed.

3.9 Marking.

3.9.1 Shell. Marking inside the helmet shell shall include the manufacturer's name or registered trademark and designation (model number). Embossing, indenting or cemented label is acceptable.

3.9.2 Headband. The range of sizes to which the headband is adjustable shall be marked on labels cemented to, or embossed upon, the helmet shell or the headband.

3.10 Workmanship. The workmanship shall be first class in every respect. The safety helmets shall have no sharp edges or burrs, and shall be free from any other defects which detract from their appearance or which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the visual and dimensional examination of 4.4.1 and the tests specified in 4.5. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.4).

4.2.1 Samples for qualification tests. Eighteen helmets of each type and material for which approval is desired shall be furnished. (Six for insulation resistance and water absorption tests, six for impact tests and six for penetration tests).

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All helmets of one type offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

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4.3.2 Sampling for visual and dimensional examination. A random sample of helmets shall be selected from each lot in accordance with MIL-STD-105 at inspection level III. The acceptable quality level shall be 2.5 defects per hundred units for major defects and 4.0 defects per hundred units for minor defects.

4.3.3 Sampling for insulation-resistance test. A random sample of helmets shall be taken from each lot in accordance with MIL-STD-105 at inspection level S-4. AQL shall be 1.5 defects per hundred units (see 4.5.2).

4.3.4 Sampling for impact, penetration, and weight tests. From each lot, a random sample of helmets shall be selected separately for each test in accordance with MIL-STD-105 at inspection level S-2. AQL shall be 1.0 defects per hundred units (see 4.5).

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on each sample helmet selected in accordance with 4.3.2 through 4.3.4. Quality conformance inspection shall consist of the following:

(a) Examination (see 4.4.1).
(b) Tests (see 4.4.2).
(c) Inspection of preparation for delivery requirements (see 4.6).

4.4.1 Visual and dimensional examination. Each of the sample helmets selected in accordance with 4.3.2 shall be visually and dimensionally examined to verify compliance with this specification. Examination shall be conducted as indicated in table I. Any helmet in the sample containing one or more visual or dimensional defects shall not be offered for delivery, and if the number of defective helmets in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

TABLE I. Classification of defects

Examine	Defect	Major	Minor
Finish	Color or type of finish not in accordance with approved sample or as required.	X	
Design	Any characteristic not in accordance with the specified requirements or approved sample seriously affecting serviceability	X	
	Any characteristic not in accordance with the specified requirements or approved sample affecting serviceability but not seriously .		X
Material (general)	Any component not fabricated of approved material seriously affecting serviceability.	X	
	Any component not fabricated of the approved material not seriously affecting serviceability, e.g., rivet is fabricated from steel when brass is specified.		X
Headbands or chinstraps	Side in contact with skin not smooth.		X
	Any sharp burr, sliver, or edge		X
	Any cuts, holes, open scratches, wrinkles, or scars affecting serviceability but not seriously.		X
	Spots or stains that are clearly noticeable .		X
	Any assembly that is not adjustable where required, e.g., chinstrap not adjustable . .	X	
	Any component or assembly malformed seriously affecting serviceability	X	

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TABLE I. Classification of defects (Cont'd)

Examine	Defects	Major	Minor
Headbands or chinstraps (con.)	Any component or assembly malformed affecting serviceability but not seriously		X
	Any fastener, i.e., eyelets, rivets missing or not securely set to a degree where it may become detached from assembly.	X	
	More than one fastener not securely set but is not expected to become detached from assembly		X
	Any fastener too tightly set or compressed resulting in damage to assembly, e.g., too much pressure applied to rivet resulting in damage to shell	X	
Assembly highlights	Attachment of the cradle assembly to helmet does not provide adequate clearance between the shell and the headband as compared with approved sample or requirement	X	
Crown straps and laces	Any rip, hole, cut, tear, or other defect seriously affecting serviceability	X	
	Defective to a degree where serviceability will be affected but not seriously, e.g., misweave, slubs, etc., if applicable		X
Shell	Any hole, soft spot, air pocket, void, evidence of delamination, rough or abraded area, cracked, chipped, blistered, pitted or otherwise defective, seriously affecting serviceability	X	
	Scratches, air pocket, void, rough or abraded area, chipped, pitted, blistered or other defective, affecting appearance but not seriously affecting serviceability . . .		X
	Metal parts and metal clips for attaching crown straps and headbands to shell, positioned in manner likely to cause injury if blow on trim forces helmet over ears or forehead.	X	
Construction and workmanship general (applicable to all components, unless otherwise indicated herein)	Any component omitted, damaged, misplaced, operation omitted or not properly performed seriously affecting serviceability	X	
	Any component misplaced, damaged, operation not properly performed affecting serviceability but not seriously.		X
	Any component not firmly affixed or attached to an extent that parts or assembly may become separated in use.	X	
	Any component not firmly affixed or attached but part or assembly is not expected to become detached in use		X

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4.4.2 Quality conformance tests. Sample helmets selected in accordance with 4.3.3 and 4.3.4 shall be subjected to the tests specified in 4.5.2 through 4.5.4.3 and 4.5.8. Sample helmets failing one or more of these tests shall not be offered for delivery. If the number of defective helmets in any sample exceeds the acceptance number for the samples tested, the lot represented by the sample shall be rejected, and all helmets returned to the contractor.

4.5 Test procedures.

4.5.1 Preparation of samples (insulation-resistance and water absorption tests). Where it is evident that the sample helmets have a protective coating over the basic material, the entire exterior surface of the shell shall be abraded until the basic material is exposed using number 60-grit garnet paper.

4.5.2 Insulation resistance test. The insulation-resistance test shall be conducted in accordance with the following paragraphs, prior to the water absorption test, to determine conformance with 3.6.1. The same samples may be used for both the insulation-resistance and water absorption tests.

4.5.2.1 Apparatus. The test apparatus shall consist of the following:

(a) A vessel, containing tapwater, of sufficient size to submerge an inverted helmet to within 1/2 inch of the junction of brim with the dome.
(b) Wire frame for suspending inverted helmet in tapwater.
(c) Source of 60-cycle alternating current with 2,200 volts (root mean square).
(d) Wiring and terminals for application of voltage across dome of helmet.
(e) Voltmeter of sufficient capacity.
(f) Milliammeter of sufficient capacity and accuracy to measure the specified current.

4.5.2.2 Procedure. An inverted helmet with the suspension removed shall be placed on the wire frame and submerged in the tapwater to within 1/2 inch of the junction of the brim with the dome. The helmet shall also contain the tapwater to the same level. If the helmet is provided with holes in the crown near the brim, the helmet shall be submerged to within 1/2 inch of the holes. One terminal from the current source shall be connected to the wire frame which supports the helmet and the other shall be immersed in the tapwater which is contained by the helmet. The 2,200 volts shall then be applied for 1 minute. If passage of current is evidenced by deflection of the voltmeter the helmet shall be considered as having failed to conform to 3.6.1. If no deflection of voltmeter occurs, a milliammeter is placed in the current and reading taken.

4.5.2.3 Report. For each specimen, the leakage current or evidence of complete breakdown shall be reported.

4.5.3 Impact test. The impact test shall be performed as follows to determine conformance with 3.6.2:

4.5.3.1 Apparatus. The test apparatus shall consist of the following:

(a) A standard head form (see 6.5). The standard head forms are of four sizes : Small, medium, large, and extra large, representing head sizes 6-3/4, 7, 7-1/4 and 7-1/2, respectively. Test-block forms which simulate the standard head forms may be employed. The blocks shall duplicate the crown dimensions and contours from the top down to the ear and eye levels, including the top portion of the ears. To protect the head form, it may be provided with a steel insert (see fig. 1). For the clearance, impact and penetration resistance tests the headband should be adjusted to fit similar to the fit obtained when worn on a man's head and the hat or cap mounted on the medium (size 7) standard head form. Where the headband cannot be adjusted to fit the medium size standard head form, the hat or cap shall be mounted on the appropriate size of standard head form.
(b) A brinell penetrator assembly as shown on figure 1, having a one inch thick steel base plate. The impression bar shall be of 1100-0 Aluminum, size 1/2 inch thick and 1-1/8 inches wide, having a predetermined brinell hardness of 21 to 24 as measured with a 500 kilogram load using a 10-millimeter ball. The brinell penetrator used in the impact test shall be a hardened-steel ball 1/2 inch in diameter (12.7 mm.).

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(c) A steel ball approximately 3-3/4 inches in diameter and weighing 7.8 to 8.0 pounds.
(d) A Brinell microscrope or other suitable micrometer microscope accurate to 0.05 millimeter (mm).

4.5.3.2 Test conditions.

4.5.3.2.1 For qualification tests. Impact tests shall be conducted with the specimens maintained at temperatures of 0° ± 2°F. and 120° ± 2°F. Specimens shall be subjected to the test temperature for at least 4 hours prior to the impact test.

NOTE: Elapsed time from removal of helmet from controlled temperature to time of impact shall not exceed 15 seconds.

4.5.3.2.2 For quality conformance tests. Impact tests shall be conducted with the specimens maintained at the prevailing atmospheric temperature.

4.5.3.3 Size of specimens (for qualification tests). Specimens shall be tested on the medium size standard head form except as noted in 4.5.3.1 (a).

4.5.3.4 Procedure.

(a) Mounting specimens. Specimens shall be properly centered on the appropriate head form, in a position similar to that when worn on a man's head, with the back of the hat or cap towards the fulcrum of the impact test base assembly.
(b) Crown clearance. The clearance shall be determined by placing the hat or cap on the appropriate size of standard head form. The crown adjustment laces, if any, shall be removed. A 25-pound weight with a flat surface of not less than 2 inches in diameter shall be placed on the shell, and the distance between the underside of the shell and the crown of the head form shall be considered to be the clearance. To determine the clearance, the shell, minus suspension, shall be placed on the head form and a dimensional reading taken as described (see fig. 2). The suspensions shall then be installed and another dimensional reading taken with the 25-pound-weight applied to the crown of the shell. The difference in dimensional readings shall be the clearance. The clearance between the shell of the specimen and the standard head form shall be not less than 1-1/4 inches minimum (measured with the tie cord out if the assembly uses a a tie cord) under a 25-pound load.
(c) Application of impact. The Brinell penetrator assembly shall be placed on a level surface that will not deflect during the test. The drop ball, the head form and the penetrator ball shall be center aligned by means of a plumb bob. For testing, the center of the crown of the hat or cap shall be as nearly centered as possible by adjustment of the headband. The hat or cap shall be mounted with the back towards the fulcrum. The clearance shall be adjusted to the minimum clearance determined in accordance with 4.5.3.4 (b). The ball shall be dropped vertically immediately after aligning the standard head form as required, from a height of 60 inches, measured from the bottom of the ball to the top of the highest point on the shell. An electromagnetic release is suitable. The ball shall not be allowed to strike the hat or cap more than once. Impressions on the bar shall be spaced at least 2-1/2 diameters apart, edge to edge. The minimum diameter of impression on the test bar shall be measured and recorded to the nearest 0.1 millimeter using a Brinell microscope. Elliptical impressions shall be disregarded if the difference between the minimum and maximum axes exceeds 0.3 millimeters. All impressions from double blows shall be disregarded. For each test, the average Brinell hardness number of the 7-inch impression bar used shall be recorded.

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4.5.3.5 Report. The average force and the greatest individual force for all helmets tested under one group shall be computed and reported. The forces shall be computed from the impression diameters using the following Brinell formula:

$$F = 2.2 \times H \frac{\pi D}{2} \times (D - \sqrt{D^2 - d^2})$$

Where:

F is the transmitted force in pounds,
H is the average Brinell hardness number of the impression bar,
D is the diameter of the impression ball in millimeters, and
d is the diameter of the impression in millimeters (see 6.6).

4.5.4 Penetration test. The penetration test shall be performed as follows to determine conformance with 3.6.3:

4.5.4.1 Apparatus. The test apparatus shall consist of a medium size standard head form or test block as specified in 4.5.3.1 (a) and a 16-ounce plumb bob with a steel point having an included angle of $35^\circ \pm 1^\circ$ and a maximum point radius of 0.010 inch.

4.5.4.2 Procedure. The specimen shall be mounted on the apparatus head form as specified in 4.5.3.4 (a) and the plumb bob dropped onto the geometric center of the crown within a 3 inch diameter circle from the height of 10 feet, measured from the bottom point of plumb bob to the top of the helmet. The depth the plumb bob has pierced the helmet shall be measured to nearest 1/32 inch and shall include the thickness of the shell material. Plumb bob shall not fall on ridges or the injection point.

4.5.4.3 Report. For each specimen, the depth of penetration to the nearest 1/32 inch shall be reported.

4.5.5 Flammability test. Method 2021 of Fed. Std. No. 406 shall be employed for the flammability test to determine conformance with 3.6.4. Three strips approximately 1/2 inch wide and 5 inches long shall be cut from the brim or sides of the crown, whichever is thinner, and shall be used in lieu of the 10 specimens required in method 2021.

4.5.5.1 Test conditions. Test specimens shall be subjected to the flammability test after subjection to the tests specified in 4.5.3 and 4.5.4.

4.5.5.2 Report. The burning rate reported shall be the average for the three specimens tested.

4.5.6 Water absorption test. Method 7031, immersion procedure "A" of Fed. Std. No. 406 shall be employed for water absorption test to determine conformance wtih 3.6.5 with the following exceptions: Three whole helmet shells prepared in accordance with 4.5.1, when necessary, shall be used in lieu of the cut specimens, and conditioning in an oven at 120°F. for 4 hours and weighed before immersion in lieu of conditioning required by method 7031. Water absorption reported shall be the average of that determined for the three specimens.

4.5.7 Headband clearance measurement. To determine conformance with 3.4.2.3, clearance between the headband and helmet shall be determined by measuring the circumference of the outside of the headband (at the maximum required size (7-1/2) for adjustable headbands in accordance with 3.4.2.2.) and of the inside of the shell at the juncture of the dome and the visor and computing the diameters and clearance by the following formulas:

$$\text{Diameter} = \frac{\text{Circumference}}{\pi}$$

$$\text{Clearance} = \frac{D-d}{2}$$

D = inside diameter of shell.
d = outside diameter of headband (size 7-1/2).

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4.5.8 Weight. Helmets shall be weighed complete without winter liner, chinstrap, nape strap or other accessories on any suitable weighing balance to the nearest tenth (0.1) ounce to determine conformance with 3.6.6.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing, and marking requirements of section 5 are complied with. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

TABLE II. Inspection of preparation for delivery.

Examine	Defect
Markings (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, improper taping, etc.
	Bulging or distortion of containers.
Content	Number per bundle is more or less than required.
Weight	Net weight of contents more than maximum specified.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.7.)

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each helmet shall be placed on a polyethylene bag fabricated from film at least 0.0015 inch thick. The bag shall be of sufficient length to enclose the helmet and permit the open end of the bag to be folded and tucked inside the helmet shell. Five wrapped helmets shall be stacked together by nesting the crowns one within the other.

5.1.2 Level C. Unless otherwise specified (see 6.2), helmets shall be packaged according to the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Each shipping container (level A or B) shall be provided with a full height slotted partition made of double-faced corrugated fiberboard which shall divide the box into four equal sections. The container and inner packing shall be designed so that a stack of five packaged helmets shall fit snugly in the partitioned section of the box.

5.2.2 Level A. Helmets packaged as specified in 5.1 shall be packed in snug-fitting wood-cleated fiberboard, wood-cleated plywood, nailed wood or wire-wound boxes conforming to PPP-B-591 (class II), PPP-B-601 (overseas type), PPP-B-621 (class 2) and PPP-B-585 (class 3), respectively. Containers shall be lined with a waterproof caseliner conforming to MIL-L-10547 sealed in accordance with the appendix thereto. Shipping containers shall be closed and strapped. The gross weight of boxes shall not exceed 150 pounds.

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5.2.3 Level B. Twenty helmets packaged as specified in 5.1 shall be packed in a close-fitting fiberboard box conforming to PPP-B-636, type CF or SF, class domestic, variety SW or DW, PPP-B-636 (Class weather resistant) or PPP-B-640 (Class 2). The box shall be closed in accordance with the appendix to the box specification.

5.2.4 Level C. Unless otherwise specified (see 6.2), helmets packaged as specified in 5.1 shall be packed in a manner which will insure acceptance and safe delivery to destination at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to marking required in the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123 (see 6.3).

5.3.2 Military agencies. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 (see 6.2).

6. NOTES

6.1 Intended use. Helmets covered by this specification are intended to protect the head and neck from being struck by falling objects or from being stuck against objects that may cause injury. This protection is especially necessary in shipbuilding and repair, steel erection, bridge, road and air field construction, mining, stevedoring, tunneling, aircraft overhaul and maintenance, building construction, timber work, and manufacturing.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2.1).
(c) If adjustable chinstrap, nape strap or winter liner are required (see 3.2 and 3.4.4).
(d) Color and color number required (see 1.2.2 and 3.5).
(e) Quantity of each required.
(f) Level of packaging, if different (see 5.1.2).
(g) Level of packing, if different (see 5.2.4).
(h) Levels of packaging and packing required (see 5.1 and 5.2).
(i) Marking required (see 5.3).

6.2.1 When suspensions or components of the suspensions (for replacement purposes) are ordered, the manufacturer's type or model number of the helmet for which the replacement suspension or component is ordered should be stated, or if required, a representative helmet should be furnished in order that the manufacturer may have the necessary identifying information.

6.2.2 For civil agency procurement. When level B packaging is required for civil agency procurement the level A requirements in 5.1.1 shall apply.

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL - GGG-H-142, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they proposed to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Department of the Navy, Center Building, Prince George's Center, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4).

6.4 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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6.5 Standard head forms. The standard head forms mentioned throughout this specification are the model heads known as "A.M.L. Head Size Standards," a set of which is located at each of the following places:

National Bureau of Standards
Washington, D.C. 20234

Aero Medical Laboratory
Wright Air Development Center
Wright-Patterson Air Force Base, Ohio 45433

Nondestructive and Tool Testing Division
Laboratory Branch
Quality Assurance Section
Philadelphia Naval Shipyard
Philadelphia, Pennsylvania 19112

6.6 For convenience, table III shows the transmitted forces for the measured impression diameters on impression bars ranging in Brinell hardness from 21 to 24 as computed from the Brinell formula (see 4.5.3.5).

TABLE III. Transmitted forces in pounds

Diameter of Impression, Millimeters	Brinell hardness number			
	21	22	23	24
3.5	455	475	495	515
3.6	480	505	525	550
3.7	505	530	555	580
3.8	535	560	590	610
3.9	565	595	620	650
4.0	600	625	655	680
4.1	630	660	690	720
4.2	660	690	720	755
4.3	695	725	760	790
4.4	725	760	795	830
4.5	760	795	835	870
4.6	795	835	870	910
4.7	A830	870	910	950
4.8	870	910	950	990
4.9	905	950	995	1035
5.0	945	990	1040	1080
5.1	B985	1035	1080	1130
5.2	1030	1080	1125	1175
5.3	1070	1120	1175	1225
5.4	1115	1165	1220	1270
5.5	1155	1210	1265	1320
5.6	1200	1255	1310	1370

NOTES: Values below line A are above the specified average force.

Values below line B are above the specified individual force.

These values give the transmitted force in pounds for impression ball 1/2 inch (12.7 mm.) in diameter.

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6.7 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

MILITARY CUSTODIANS:

Army - GL
Navy - SH
Air Force - 82

Review activities:

Army - GL
Navy - SA, SH, YD
Air Force - 82
DSAH - DP

User activity:

Army - CE

Preparing activity:

Navy - SH

CIVIL AGENCIES INTEREST:

GSA-FSS
INT-BR
TR-CG
LABOR

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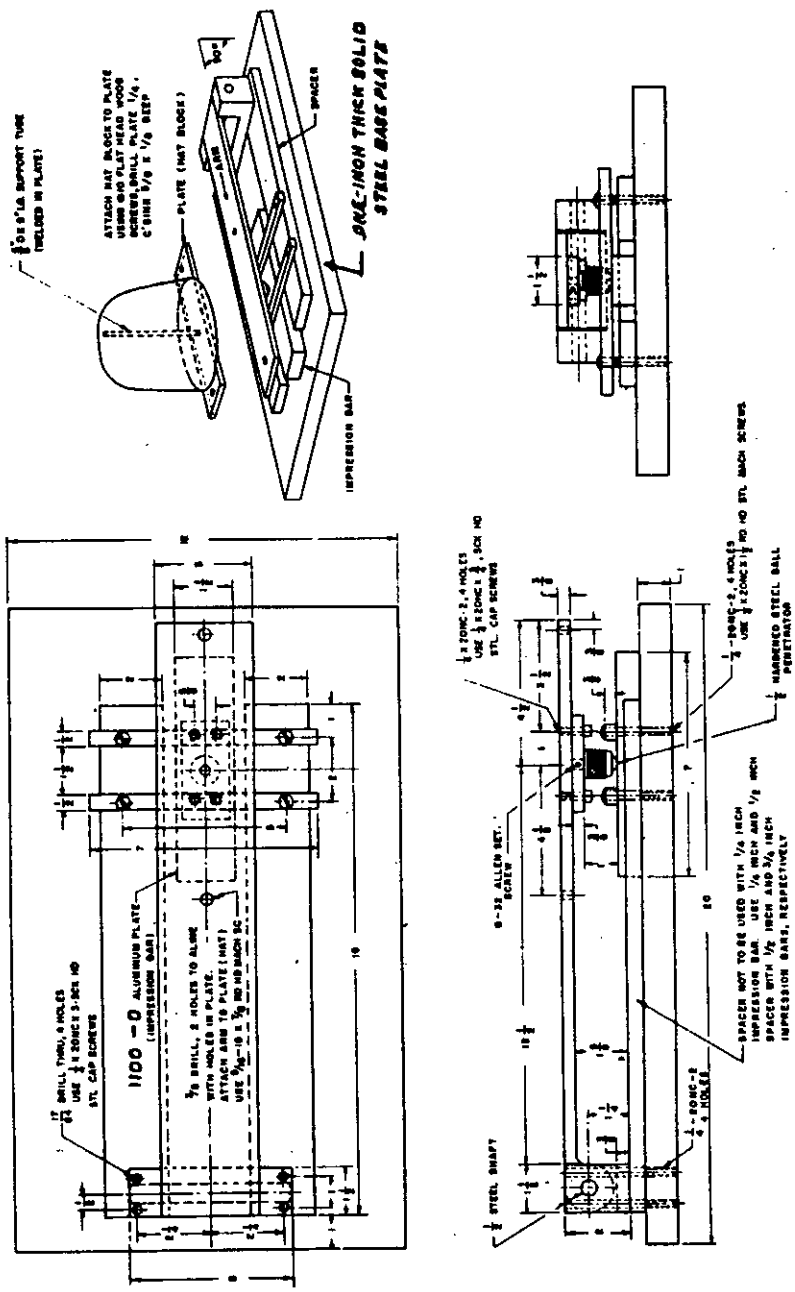


Figure 1. - Brinell hardness penetrator assembly

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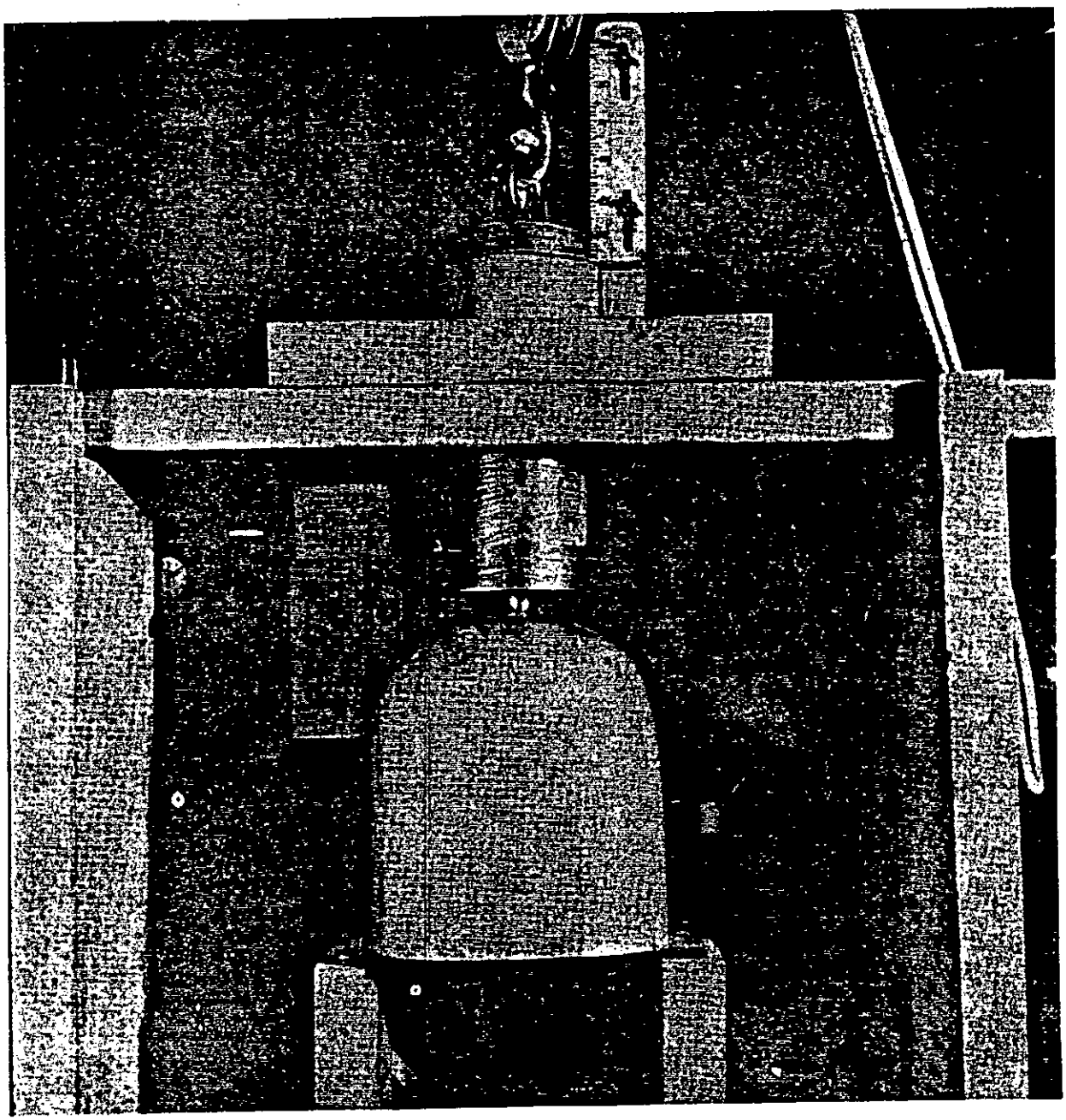


Figure 2. Suggested apparatus for measurement of crown clearance (see 4.5.3.4 (b)).

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:

5. *PROBLEM AREAS*

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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